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Massive open online courses learners' continuance intention: shaping a roadmap to micro-credentials

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Abstract

Purpose – This study intends to investigate the determinants of learners' continuance intention to use massive open online courses (MOOCs) for personal or professional development.

Design/methodology/approach – This study employed quantitative research design. The respondents were individual learners from six selected universities in China who used MOOCs for continuous learning. A purposive sampling technique was employed to obtain 270 valid samples. Data were analyzed and analytical outputs were produced using the techniques of Partial Least Squares Structural Equation Modeling and Importance-Performance Matrix.

Findings – Expectation confirmation was found to have a positive relationship with perceived usefulness, flow experience, learning self-efficacy and satisfaction with MOOCs. Perceived usefulness, flow experience and leaning self-efficacy were also found to have a positive relationship with MOOC satisfaction. In addition, perceived usefulness, flow experience, learning self-efficacy and MOOC satisfaction had a positive impact on continuance usage intention.

Originality/value – The outcomes of the study can serve as a practical reference for MOOC providers and decision-makers to develop relevant strategies to increase the course completion rates.

Keywords Expectation–confirmation theory, Flow experience, Self-efficacy, Satisfaction, Perceived usefulness, Continuance intention behavior, Micro-credential

Paper type Research paper

1. Introduction

Massive open online courses (MOOCs) consist of a wide range of courses offered to learners world-wide via a web-based open learning approach which caters to the needs of individuals looking for personal growth and professional development for a minimum fee (Chaw and Tang, 2019a). A unique and appealing feature of this learning platform is that it provides learners with flexibility and autonomy in their learning with regards to course selection, timing and pace (Wong, 2021). This unique feature has sparked a great interest among the prominent Chinese institutions to join the international MOOC platforms (Lu *et al.*, 2018). According to Wu Yan, the director of Ministry of Education in China, over 460 colleges and universities in China have launched more than 10 MOOC platforms with more than 3,200 MOOCs (Liangyu, 2018). Fifty-five million students in China have used MOOCs and more than six million have earned MOOC credits at the university level (Li and Liu, 2018). In the beginning of 2022, there were 24 MOOC platforms providing more than 69,000 MOOCs in

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Chinese language, about double the number in 2020 (Ma, 2022). These activities have demonstrated China's tremendous progress in massive learning via the internet platform.

MOOCs are often affiliated with educational institutions; for example, Stanford University, Harvard University and MIT in the US; and Tsinghua University, Shanghai Jiao Tong University and Fudan University in China. However, the course completion rate of MOOCs has always been low, with a range between 2 and 13% (Chaw and Tang, 2018; Shao, 2018). This low completion rate is an issue for MOOC providers around the world that needs attention, especially given the opportunity for self-learning in the context of Covid-19 pandemic. Furthermore, MOOC is a new learning innovation; there is a lack of research exploring the critical reasons or factors behind the low continuance usage intention toward MOOCs, as compared to other technology-based learning platforms.

There is another online learning platform similar to MOOC and is called micro-credential. Micro-credential offers courses in smaller units or modules to upskill the learners with knowledge, values and competencies in a narrow area of study or practice (Shah, 2022). Micro-credential is mainly offered via industry and third-party collaborations where learners will attain evidence-rich credentials (Kumar *et al.*, 2022). In a similar vein, micro-credential is also experiencing a low completion rate (Burke, 2019). Burke (2019) reported that "The apparently low programme completion rates likely reflect that most learners do not perceive the benefits to outweigh the costs." Nevertheless, the focus of this study is on MOOCs mainly offered by universities. The MOOC platforms would foster the development of micro-credentials given their similarities with MOOCs (Wheelahan and Moodie, 2021). The examination of learners' continuance intention toward MOOCs will provide useful information to the micro-credential practitioners.

The purpose of this research is to determine whether factors such as expectation confirmation, perceived usefulness, satisfaction, flow experience and learning self-efficacy would influence leaners' continuance intention to use MOOCs in China. The results of the current study would have implications on addressing the issue of high drop-out rate of MOOCs. Ho (2010) adopted a combined model of expectation confirmation model (ECM), technology acceptance model (TAM), cognitive model (COGM) and self-determination model (SDM) in studying the constructs affecting the continuance usage intention toward e-learning. The author found that motivation (autonomy, competence, relatedness), belief (confirmation, perceived usefulness, perceived ease of use) and attitude (satisfaction) all have excellent explanatory power in relation to continuance intention. Lee (2010) integrated ECM, TAM, theory of planned behavior (TPB) and flow theory to explain and predict users' continuance intention toward e-learning and found that satisfaction has the greatest significant effect on learners' continuance intention, followed by perceived usefulness. Similarly, Ashrafi *et al.* (2020) used the combined model of expectation confirmation theory (ECT) and TAM, and the outcomes of the study disclose that perceived usefulness is the strongest predictor of students' continuance intention to use LMS. Zhou (2017) explored the factors influencing leaners' continuance usage intention toward MOOCs in China using ECM and found that the predicting constructs (i.e. confirmation of prior learning experience, satisfaction with prior learning experience and perceived usefulness) were supported by the empirical data. Lu *et al.* (2019) combined ECM with flow theory and validated expectation confirmation's positive influence on flow and flow's positive influence on users' satisfaction. Furthermore, their study indicated that satisfaction is a significant factor to determine users' behavior in terms of continuing using MOOC. Through a modified ECM, Gu *et al.* (2021) integrated ECM and the quality factors of Delone and McLean IS success model and found that three critical antecedents (information quality, system quality and service quality) affect users' confirmation expectation for a MOOC platform. Among these three antecedents, service quality has the most impact on learners' expectation confirmation. Shang and Lyv (2022) extended ECM by integrating task-technology fit model (TTF) and flow theory to investigate the impact of quality elements on continuance intention. The findings revealed

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that teaching-based quality can boost learners' TTF, and confirmation and platform-based quality can increase confirmation and the perceived value of learning MOOCs.

Through the selective review of the previous studies, it can be seen that most of the past studies from the perspective of ECM examine continuance intention on e-learning platforms by combining ECM and other theories such as TAM, TTF and flow theory. To the best of the authors' knowledge, no study has combined flow theory and self-efficacy theory to further extend ECM model in examining the e-learning platform. The influence of flow experience and self-efficacy in MOOCs cannot be ignored because flow experience and self-efficacy are two inner elements related to learners' motivation in adopting e-learning. By introducing flow theory and self-efficacy theory, we can further complement the fundamental model of ECM. Hence, the objective of the present study is to examine whether expectation confirmation, perceived usefulness, satisfaction, flow experience and learning self-efficacy have an influence over learners' continuance usage intention toward MOOCs.

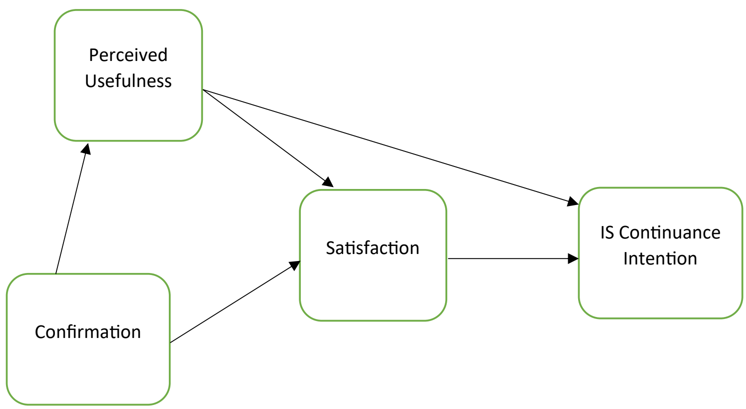
The following sections cover the theories involved, hypothesis and research model development, research methods, data analysis and the subsequent discussion of the research findings along with future research directions.

2. Theoretical background

2.1 Expectation-confirmation model (ECM)

ECM originated from the expectation confirmation theory (ECT) proposed by Oliver (1980). Initially as a marketing theory, ECT assumes that consumer's expectation, along with perceived performance, brings about consumer's satisfaction after purchase (Oliver, 1980; Spreng *et al.*, 1996). Consumer's confirmation of previous purchase experience just mediates this influence. In other words, consumers will continue to buy the product or service when they positively confirm their expectations. Likewise, they will stop buying the product or service when they are not able to confirm their expectations based on their previous consumption experience.

Bhattacherjee (2001) first applied ECT to examine continuance usage intention in the field of information system and tested an ECM of information system as shown in Figure 1. In this model, expectation confirmation affects users' continuance usage intention via both users' perceived usefulness and satisfaction. Perceived usefulness, on the other hand, has both direct and indirect effect on users' continuance usage intention, with users' satisfaction as the mediator. ECM has expanded into the field of e-learning over the recent years. Lee (2010) integrated ECM, TAM, the theory of planned action (TPA) and the flow theory to explain



Source(s): Bhattacherjee (2001)

Figure 1.
ECM-IS model

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users' continuance intention toward e-learning. Dai *et al.* (2020) enhanced the extended ECM model by incorporating affective and cognitive variables that encompass reflections on the past and expectations for the future, as well as considering intrinsic and extrinsic motives, to elucidate the intention of learners to persist in MOOC learning.

2.2 Flow Theory

Flow is described as "the holistic feeling that people have when they act with entire participation," meaning a state wherein an individual is fully absorbed in the experience being done for the moment, without any interruptions, whatever action is being performed (Csikszentmihalyi, 1997). Flow experience has been recognized as a phenomenon that can be experienced by almost everybody, regardless of the action's qualities or cultural differences (Özkara, 2015). Research studies on flow experience have indicated that people may feel "the flow" in a variety of activities, for example playing games, reading and watching films (Pelet *et al.*, 2017). Chen *et al.* (2022) highlight that the use of information technology usually happens with a flow experience, which influences user satisfaction and continue use behaviors. When learners are in the flow state, they will work at their highest potential and make more progress in their academic learning in MOOCs (Hong *et al.*, 2016). The flow experience has three antecedents: first, anticipated obstacles of the flow activity that match the participants' capacity; second, defined goals for the flow activity; and, third, timely feedback on how well individuals have done in attaining their goals (Nakamura and Csikszentmihalyi, 2009).

It is worth noting that, while flow appears to be a comparable emotion for everyone, there have been several differences over the structure of flow (Novak *et al.*, 2000). Perceived enjoyment, perceived control and attention are three notions that Koufaris (2002) devised to quantify flow. However, Huang (2003) incorporated four components: control, attention focus, curiosity and intrinsic interest to examine flow. Similarly, according to Li and Browne (2006), flow experience comprises four dimensions: focused attention, control, curiosity and temporal dissociation. Flow is typically believed to have qualities such as sensation of control, interest, time distortion, attention and perceived satisfaction, but the lack of clarity on flow's antecedents and subdimensions has been a major problem (Özkara *et al.*, 2016). Lee (2010) combined ECM, TAM, TPB and the flow theory to hypothesize a theoretical model to explain and predict users' intentions to continue using e-learning, with flow being conceptualized into two subdimensions of concentration and perceived enjoyment. However, rather than broadening flow's original meaning by incorporating its various subdimensions, flow experience is a unidimensional concept in the present study in examining MOOC learners' continuance usage intention in China.

2.3 Self-efficacy theory

Albert Bandura (1977), a psychologist and professor at Stanford University, coined the concept of self-efficacy. The conviction in one's "capabilities to plan and execute the courses of action necessary to generate specific attainments" is referred to as one's perceived self-efficacy (Bandura, 1997, p. 3). In the context of MOOC learning, self-efficacy has been referred to as computer self-efficacy, technological self-efficacy, academic self-efficacy and learner self-efficacy (Fatima *et al.*, 2022). In general, computer self-efficacy is significant in molding an individual's feelings and behavior (Compeau and Higgins, 1995). It influences an individual's expectations about the results of using computers and, as a result, influences his or her decision to use computers (Hill *et al.*, 1987).

Technological self-efficacy pertains to learners' beliefs regarding their ability to utilize technology tools and platforms to engage in learning activities and achieve the intended learning objectives (Keengwe, 2007). Researchers have discovered that technical self-efficacy has a considerable impact on technology acceptance and use (Celik and Yesilyurt, 2013). For instance, the level of students' technical self-efficacy significantly impacts their intention toward utilizing technological tools and their evaluations of technology's effectiveness in facilitating learning.

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The motivation of students to engage in technological learning is evident through their acquisition and familiarity with technical skills during the process of self-directed learning with technology. The extent of technological learning motivation has a notable impact on students' intention to utilize online learning platforms and other technology-related resources. Students with stronger technical self-efficacy have been found to be more satisfied with their online learning experience. Improving students' technology self-efficacy in an online learning environment can make them pleased with their learning and obtain better scores (Mew and Honey, 2010; Wang *et al.*, 2013).

3. Hypotheses development

3.1 Expectation confirmation

Users' perception of the congruence between their expectations of information system uses and its real performance is defined as expectation confirmation by Bhattacherjee (2001). In a study conducted by Jo (2022), it was found that the confirmation of expectations significantly influences the satisfaction of students taking online classes. Likewise, numerous studies on e-learning have supported the positive association between users' confirmation and their satisfaction (Wang *et al.*, 2021; Gu *et al.*, 2021; Shang and Lyv, 2022). Users' confirmation will be positive if they deem that their actual performance exceeds their earlier expectations. On the other hand, users' expectation confirmation will be negative when they believe their actual performance is poorer than their earlier anticipation, and, consequently, users' satisfaction will be lowered (Peng *et al.*, 2019; Samir *et al.*, 2020). Hence, when learners' real performance surpasses their prior expectations on MOOC platforms, they would feel pleased with their MOOC learning. The following hypothesis is then proposed:

H1. Learners' expectation confirmation of prior learning experience has a positive influence on their satisfaction with MOOC learning.

In ECM, perceived usefulness refers to users' expectations for the future once they have used something for the first time (Bhattacherjee, 2001). The degree of their past usage experience is positively associated with their post-adoption anticipation (Lee, 2010). When their previous expectations are met, they will believe that the system is beneficial to use and will anticipate more from it in the future. The finding of a study conducted by Zhao and Hu (2022) on MOOC showed that learners' perceived usefulness of this learning platform can be significantly influenced by expectation confirmation. Hence, learners' validation of earlier learning experiences would boost their expectations for future learning. MOOC platforms generally have a solid reputation among learners since they are backed by well-known universities or lecturers (Alraimi *et al.*, 2015). Students can use MOOCs to broaden their knowledge on certain topics or areas (Hew and Cheung, 2014). Therefore, we proposed the following hypothesis:

H2. Learners' expectation confirmation of prior learning experience has a positive influence on their perceived usefulness of MOOC learning.

Flow experience here refers to the optimal and happy experience from the complete engagement in and concentration on MOOCs learning (Lee, 2010). When the learners' expectation is confirmed, they tend to have a greater eagerness to use MOOCs and become more committed with a flow experience. Lu *et al.* (2019) work showed a positive influence of expectation confirmation on flow, which, in turn, positively influenced intention to continue using MOOCs. Shang and Lyv's study (2022) indicated that confirmation of expectation has a significant influence on users' flow experience in MOOCs. Mulik *et al.* (2019), in their study of undergraduate students' continuance usage intention toward online learning platforms, also revealed that expectation confirmation has a positive and significant impact on learners' flow experience. We hypothesize that:

H3. Learners' expectation confirmation of prior learning experience has a positive effect on their flow experience when using MOOCs.

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Learning self-efficacy refers to learners' self-determination and self-confidence that they will be able to achieve their academic goals in MOOCs context (D'Souza *et al.*, 2023). Past studies (Shiau *et al.*, 2020; Susanto *et al.*, 2016) have shown the influential roles of users' expectation confirmation and self-efficacy on continuous intention to use a product or service. Moreover, Peaslee (2018) has established a correlation between faculty members' confirmation behaviors and students' learning self-efficacy. When learners confirm their expectations positively, they tend to have greater motivation and confidence in themselves. Subsequently, they believe they can continue learning MOOCs to accomplish their goals (Malik and Rao, 2019). Hence, it is postulated that:

H4. Learners' expectation confirmation of prior learning experience has a positive influence on their learning self-efficacy in MOOCs.

3.2 Perceived usefulness

The concept of perceived usefulness is significant in ECM because it indicates the usefulness of using information systems (Bhattacherjee, 2001). In the context of MOOCs, perceived usefulness refers to a learner's opinion that using MOOCs will improve their job/learning performance (Venkatesh *et al.*, 2003). Unlike traditional classrooms, e-learning takes place solely online and lacks nonverbal cues such as eye contact, movement and body movements (Proserpio and Magni, 2012). This distinction encourages scholars to examine the outcomes of e-learning. In the context of e-learning, perceived usefulness has been widely acknowledged to have a positive and significant impact on learners' satisfaction (Gu *et al.*, 2021; Ashrafi *et al.*, 2020; Wang *et al.*, 2021). Additionally, perceived usefulness has also been found to have a positive and significant impact on learners' continuance usage intention by Lu *et al.* (2019) and Mo *et al.* (2021). Hence, the following two hypotheses are proposed as per literature:

H5. Learners' perceived usefulness positively influences their satisfaction with MOOC learning.

H6. Learners' perceived usefulness positively influences their continuance usage intention of MOOCs.

3.3 Satisfaction

Satisfaction is a very significant concept in ECM. Users' desire to continue using information system is strongly influenced by two factors: their satisfaction with their previous use experience and post-adoption anticipation of future use (Bhattacherjee, 2001). Based on previous studies, a strong and positive connection has been found between users' satisfaction and their expectation to use a certain information system in the future (Alraimi *et al.*, 2015; Lee, 2010). Learners' perceptions of enjoyment and accomplishment in the learning environment are referred to as satisfaction with MOOCs (Alraimi *et al.*, 2015). In e-learning, learners' satisfaction with prior learning experience has a positive influence on their continuance usage intention as shown in the previous research studies by Liu *et al.* (2020), Gu *et al.* (2021) and Wang *et al.* (2021). In other words, learners are more likely to continue using MOOCs in the future, when they feel satisfied with MOOCs. Hence, we suggest the hypothesis:

H7. Learners' satisfaction has a positive influence on their continuance usage intention of MOOCs.

3.4 Flow experience

Flow has a considerable effect on users' contentment, according to Bakker (2008), implying that this ideal psychological state is a fundamental antecedent of users' happiness. As

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highlighted by Farhat *et al.* (2022), positive outcomes can be achieved in education, IT acceptance and the widespread adoption of the online environment through flow positive experience. Cheng (2014) used ECM in combination with quality and flow to investigate nurses' intention to continue using blended e-learning. The author found that system quality, information quality, support service quality and instructor quality have played a key role in influencing perceived usefulness, confirmation and flow; and all these factors impacted nurses' satisfaction and intention to continue using the system. Guo *et al.* (2016) empirically constructed a comprehensive model that investigates the factors that influence students' flow experience in e-learning and how it relates to their intention to continue using the platform. The authors found that telepresence is the most important factor in influencing students' flow experience, and that perceived hedonic value is the most important mediator in transmitting the effects of flow to learners' continuance usage intention. Fang *et al.*'s (2019) study demonstrated that learners' immersive experience in MOOCs has a positive and significant impact on their satisfaction of competence needs, relatedness needs and autonomy needs. These satisfaction levels, in turn, have an influence on learners' engagement in MOOC learning. However, when Mulik *et al.* (2019) incorporated flow into ECM in examining college students' long-term usage intentions for online learning platforms, flow was found to have a significant impact on learners' long-term usage intentions but not with learners' satisfaction. The research of Wang and Lin (2021) demonstrated that flow experience is a crucial factor for affecting the learners' satisfaction and continuance intention to use mobile learning applications. Shang and Lyv (2022) found that flow is positively correlated with learners' satisfaction in MOOCs; subsequently, satisfaction can positively influence learners' continuance usage intention. Similarly, the study conducted by Zhao and Khan (2022) revealed a significant correlation between students' flow experience and their perceptions of usefulness, satisfaction and intention to continue using a particular system or platform. Lastly, Gao (2023) conducted a study on smart education and discovered a positive impact of flow on continuance intention. Thus, the following hypotheses are suggested:

H8. Learners' flow experience has a positive influence on their satisfaction with MOOC learning.

H9. Learners' flow experience has a positive influence on their continuance usage intention of MOOCS.

3.5 Learning self-efficacy

Learning self-efficacy here refers to learner's internal self-confidence or self-determination to complete the online courses or programs. Heng *et al.* (2020) found a positive and significant relationship between learning self-efficacy and both satisfaction and continuance usage intention toward MOOCs. Similarly, Kuo *et al.* (2021) found a significant correlation between Internet learning self-efficacy and online learning engagement. In a separate study conducted by Aslam *et al.* (2022), it was found that both smartphone self-efficacy and the behavioral intention to use smartphones have a positive impact on students' academic performance. Saeed *et al.* (2020) also examined technology self-efficacy and found a positive relationship with continuance intention to use e-learning platforms. However, Liu *et al.*'s (2020) study did not support the hypothesis of the positive relationship between users' self-efficacy in the virtue learning community and their continuance usage intention. Therefore, to further investigate the impact of learning self-efficacy on satisfaction and continuance usage intention toward MOOCs, the following hypotheses are postulated:

H10. Learners' learning self-efficacy has a positive influence on their satisfaction with MOOCs learning.

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H11. Learners' learning self-efficacy has a positive influence on their continuance usage intention of MOOCs.

3.6 Research model

The research model that integrates ECM, flow theory and self-efficacy theory is depicted in Figure 2.

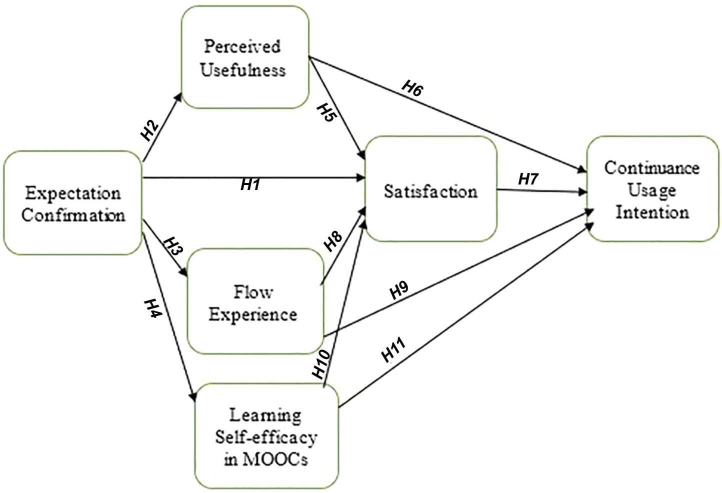
4. Research methods

4.1 Instrument development

As present study uses quantitative approach, a quantitative survey was employed to gain insightful information regarding learners' continuance intention to adopt MOOCs. A pre-test was conducted by inviting two academic experts in MOOCs to review the questionnaire. Feedback was provided on language ambiguity, and the questionnaire was subsequently amended accordingly. Additionally, we had recruited 10 students to fill up the questionnaire to ensure the questionnaire had no unfamiliar references or obscured terminology in them. This process was completed before conducting a survey. The questionnaire was divided into two main sections. The first section included respondents' demographic details such as gender, age, education level and the duration of using MOOCs. The second section of the questionnaire consisted of the influential factors of learners' intention to continue pursuing MOOCs (i.e. expectation confirmation, perceived usefulness, learners' satisfaction, flow experience, learning self-efficacy and continuance usage intention). The items measuring these variables were derived and adapted from the previous literature to ensure content validity. In the model, each of the main constructs was measured using a 5-point Likert scale ranging from "1" (strongly disagree) to "5" (strongly agree). Four items were used to measure each construct, yielding a total of 24 items included in the questionnaire. Refer Table 1 for measurement items and their sources.

4.2 Statistical power

Power analytics of Cohen (1988) was used to determine the minimum sample size of the respondents. Based on the maximum predictors of four, effect size of 0.15 and 80% power, the



Source(s): Author's own work

Figure 2. Proposed research model

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minimum sample size identified was 85. However, 316 sets of questionnaires were collected from the respondents. After filtering for straight lining pattern of the data, 270 sets of questionnaires were eligible for the data analysis. which is the minimum required sample size.

4.3 Participants and setting

Due to the unavailability of a sampling frame, the researchers adopted a non-probability sampling technique for their study. We used purposive sampling technique to obtain information from specific groups of people (Sekaran and Bougie, 2013). Firstly, only respondents who are currently or previously MOOC learners were invited to participate in the online survey. Secondly, an online questionnaire link was forwarded to students of the selected six universities in China who were briefed about the purpose of the survey. The utilization of purposive sampling technique in similar situations or contexts has been widely adopted (Iva Adeline *et al.*, 2023; Shi-Hui *et al.*, 2022).

4.4 Data collection

The data were collected from individuals pursuing MOOCs from the four cities in China, namely Shanghai, Zhengzhou, Xi'an and Changchun, which have major groups of learners and education institutions. Respondents were informed about the aim of the study. Additionally, the survey form indicated that the respondents' identity would remain anonymous and confidential. The total duration of the data collection lasted three months, from March to June 2022. The staff working in the universities posted the questionnaire in the alumni WeChat Group to encourage more respondents to fill in the questionnaire, and thus the response rate cannot be determined.

4.5 Controlling common method bias (CMB)

The common method bias was checked using the procedural remedy of Podsakoff *et al.* (2003). Ten students who had MOOC learning experience were invited to fill up the online questionnaire to identify any confusing or ambiguous questions and then modify them accordingly. In addition, the Harman single factor of statistical remedy showed the absence of common method bias as the score was 44.3%.

4.6 Analytical method

Partial least squares structural equation modeling (PLS-SEM) was employed to analyze the data using SmartPLS 4.0 software. The composite nature of the social science research renders PLS-SEM is an appropriate method for analysis. PLS-SEM can generate synergistic results via higher-order constructs due to its complex configuration approach (Ringle *et al.*, 2020). Both the convergent reliability and discriminant validity of the data were assessed

Table 1. Measurement items and sources

Constructs	No. of items	Sources
Expectation confirmation (CF)	4	Bhattacherjee (2001), Gu *et al.* (2021)
Satisfaction (SA)	4	Bhattacherjee (2001), Wan *et al.* (2020)
Perceived usefulness (PU)	4	Gu *et al.* (2021), Tao *et al.* (2019)
Flow experience (F)	4	Shang and Lyv (2022)
Learning self-efficacy (SE)	4	Bhattacherjee *et al.* (2008), Aldhahi *et al.* (2021)
Continuance usage intention (CUI)	4	Bhattacherjee (2001), Li and Zhao (2021)

Source(s): It is based on the sources in Table 1

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using the measurement model, whereas the hypothesized relationships were examined using the structural model (Chaw and Tang, 2019b, 2023; Al-Mekhlafi *et al.*, 2022; Ali *et al.*, 2022a, b).

5. Data analysis and results

Table 2 exhibits the demographic information of the respondents. Of the 270 respondents, 140 respondents (51.9%) were female, and 130 respondents (48.1%) were male. About one-third of the learners were between 20 and 25 years old (36.7%), followed by 31 and above (24.4%), between 26–30 (23.7%) and below 20 (15.2%). About half of the respondents owned a bachelor's degree (55.9%), followed by a master's degree (21.9%), diploma holders (17.8%) and the remaining were PhD holders (4.4%). Lastly, about one-third of the respondents have used MOOCs for learning for a period of 3–6 months (29.3%), followed by a period of below 3 months (25.9%), between 6 months and 1 year (25.2%) and more than 1 year (19.6%).

5.1 Measurement model

Table 3 shows that the composite reliability (CR) values are more than the 0.70 threshold value, while the average variance extracted (AVE) values are greater than the 0.50 threshold value. As seen in the table, all AVE values are more than 0.50; item loading values of equal or greater than 0.50 are acceptable with the summation of loadings contributing to a high loading score (Byrne, 2016). Fornell-Larcker criterion was performed to verify the discriminant validity. Table 4 depicts that the correlations between a construct and the other constructs in the model are smaller than the square root of AVE of that construct; therefore, discriminant validity has been established. Table 5 exhibits the results of heterotrait-monotrait ratio of correlations (HTMT). All the values have met the HTMT 0.85 criterion, and therefore, discriminant validity is established (Gold *et al.*, 2001). Table 6 demonstrates the results of cross loadings of all indicators. All the indicators show that own constructs are loaded high compared to other constructs that are loaded low. Thus, discriminant validity has also been achieved.

The evaluation of the variance inflation factor (VIF) serves as the starting point for the structural model assessment. The VIFs ranged between 2.440 and 3.022, which were lower than the 3.3 threshold value. Consequently, there was no multicollinearity issue detected in the data set. The R^2 for the targeted variables of continuance usage intention, flow experience,

		Frequency	Valid percent	Cumulative percent
Gender	Male	130	48.1	51.9
	Female	140	51.9	100.0
Age	Below 20	41	15.2	15.2
	20–25	99	36.7	51.9
	26–30	64	23.7	75.6
	31 and above	66	24.4	100.0
Education level	Diploma	48	17.8	17.8
	Bachelor's degree	151	55.9	73.7
	Master's degree	59	21.9	95.6
	PhD	12	4.4	100.0
Duration of using MOOCs	Below 3 months	70	25.9	25.9
	3–6 months	79	29.3	55.2
	6 months–1 year	68	25.2	80.4
	More than 1 year	53	19.6	100.0

Source(s): Created by authors

Table 2. Respondents' demographics

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Constructs	Indicators	Item loadings	CR	AVE
Expectation confirmation	EC1	0.745	0.868	0.622
	EC2	0.837		
	EC3	0.752		
	EC4	0.816		
Flow experience	F1	0.817	0.857	0.600
	F2	0.675		
	F3	0.797		
	F4	0.802		
Perceived usefulness	PU1	0.808	0.872	0.630
	PU2	0.773		
	PU3	0.785		
	PU4	0.810		
Learners' satisfaction	SA1	0.771	0.862	0.609
	SA2	0.751		
	SA3	0.786		
	SA4	0.813		
Learning self-efficacy in MOOCs	SE1	0.801	0.854	0.595
	SE2	0.718		
	SE3	0.788		
	SE4	0.775		
Continuance usage intention	CUI1	0.753	0.872	0.630
	CUI2	0.812		
	CUI3	0.828		
	CUI4	0.780		

Source(s): Created by authors

Table 3.
The measurement model: examination of convergent validity

	Continuance usage intention	Expectation confirmation	Flow experience	Perceived usefulness	Learners' satisfaction	Learning self-efficacy in MOOCs
Continuance usage intention	0.794					
Expectation confirmation	0.669	0.789				
Flow experience	0.736	0.719	0.775			
Perceived usefulness	0.684	0.746	0.625	0.794		
Learners' satisfaction	0.676	0.785	0.673	0.729	0.781	
Learning self-efficacy in MOOCs	0.776	0.696	0.774	0.664	0.671	0.771

Source(s): Created by authors

Table 4.
Examination of discriminant validity: Fornell-Larcker criterion

perceived usefulness, satisfaction and learning self-efficacy in MOOCs were 0.685, 0.517, 0.557, 0.680 and 0.484, respectively. These R^2 values indicate that the explanatory variables explained 68.5%, 51.7%, 55.7%, 68.0% and 48.4% of the variation in the targeted variables. The standardized root mean square residual (SRMR) has a value of 0.078, which was less than the 0.08 cut-off value, indicating a satisfactory model fit.

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	Continuance usage intention	Expectation confirmation	Flow experience	Perceived usefulness	Learners' satisfaction	Learning self-efficacy in MOOCs
Continuance usage intention						
Expectation confirmation	0.708					
Flow experience	0.638	0.720				
Perceived usefulness	0.634	0.747	0.537			
Learners' satisfaction	0.666	0.794	0.628	0.711		
Learning self-efficacy in MOOCs	0.707	0.696	0.501	0.519	0.607	

Source(s): Created by authors

Table 5. Examination of discriminant validity: heterotrait-monotrait ratio of correlations (HTMT)

Indicators	Continuance usage intention	Expectation confirmation	Flow experience	Perceived usefulness	Learners' satisfaction	Learning self-efficacy in MOOCs
CUI1	*0.753*	0.545	0.534	0.537	0.549	0.614
CUI2	*0.812*	0.506	0.602	0.480	0.483	0.639
CUI3	*0.828*	0.557	0.619	0.578	0.595	0.611
CUI4	*0.780*	0.514	0.578	0.573	0.517	0.599
EC1	0.517	*0.745*	0.517	0.524	0.562	0.482
EC2	0.516	*0.837*	0.605	0.616	0.656	0.549
EC3	0.556	*0.752*	0.571	0.574	0.593	0.573
EC4	0.523	*0.816*	0.572	0.632	0.660	0.586
F1	0.584	0.613	*0.817*	0.560	0.561	0.651
F2	0.471	0.439	*0.675*	0.366	0.391	0.462
F3	0.621	0.592	*0.797*	0.470	0.530	0.645
F4	0.592	0.567	*0.802*	0.521	0.584	0.619
PU1	0.562	0.628	0.565	*0.808*	0.611	0.565
PU2	0.480	0.531	0.458	*0.773*	0.547	0.478
PU3	0.528	0.549	0.442	*0.785*	0.569	0.507
PU4	0.592	0.651	0.512	*0.810*	0.585	0.553
SA1	0.532	0.577	0.542	0.556	*0.771*	0.532
SA2	0.472	0.585	0.431	0.605	*0.751*	0.485
SA3	0.507	0.614	0.539	0.493	*0.786*	0.497
SA4	0.591	0.670	0.582	0.620	*0.813*	0.574
SE1	0.618	0.595	0.647	0.588	0.555	*0.801*
SE2	0.535	0.534	0.514	0.509	0.535	*0.718*
SE3	0.622	0.510	0.652	0.478	0.494	*0.788*
SE4	0.615	0.502	0.570	0.467	0.482	*0.775*

Note(s): The italics showed that the parent constructs' cross loadings values are higher than other constructs and therefore discriminant validity is confirmed
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Table 6. Examination of discriminant validity: cross loadings

The path coefficient analysis in both Table 7 and Figure 3 shows that the relationships between expectation confirmation and the targeted variables of flow experience (β = 0.719,

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	Beta coefficient	T-statistics	p-value	VIF	f^2	R^2	Q^2
Expectation confirmation → flow experience	0.719	20.153	0.000	1.000	1.071	0.517	0.561
Expectation confirmation → perceived usefulness	0.746	24.362	0.000	1.000	1.255	0.557	0.549
Expectation confirmation → learners' satisfaction	0.430	6.815	0.000	3.022	0.191	0.680	0.495
Expectation confirmation → Learning self-efficacy in MOOCs	0.696	19.202	0.000	1.000	0.939	0.484	0.583
Flow experience → learners' satisfaction	0.117	1.986	0.026	2.966	0.015		
Perceived usefulness → learners' satisfaction	0.266	4.308	0.000	2.486	0.089		
Learning self-efficacy in MOOCs → Learners' satisfaction	0.104	1.703	0.046	2.945	0.012		
Flow experience → continuance usage intention	0.240	3.964	0.000	2.817	0.065	0.685	0.443
Perceived usefulness → continuance usage intention	0.195	3.559	0.000	2.440	0.050		
Satisfaction → continuance usage intention	0.115	2.077	0.019	2.626	0.016		
Learning self-efficacy in MOOCs → continuance usage intention	0.383	5.892	0.000	2.953	0.158		

Table 7.
Relationship testing

$p < 0.05$), perceived usefulness ($\beta = 0.746, p < 0.05$), learners' satisfaction ($\beta = 0.430, p < 0.05$) and learning self-efficacy in MOOCs ($\beta = 0.696$, $p < 0.05$) are significant. Flow experience ($\beta = 0.117, p < 0.05$), perceived usefulness ($\beta = 0.266, p < 0.05$) and learning self-efficacy in MOOCs ($\beta = 0.104$, $p < 0.05$) show a significant and positive impact on satisfaction. In addition, flow experience ($\beta = 0.240$, $p < 0.05$), perceived usefulness ($\beta = 0.195$, $p < 0.05$), learners' satisfaction ($\beta = 0.115$, $p < 0.05$) and learning self-efficacy in MOOCs ($\beta = 0.383$, $p < 0.05$) also positively and significantly related to continuance usage intention.

For the effect size (f^2) analysis, expectation confirmation showed a substantial effect on flow experience, perceived usefulness and learning self-efficacy in MOOCs. Expectation confirmation had a medium effect on learners' satisfaction. Learning self-efficacy in MOOCs also had a medium effect on continuance usage intention. Perceived usefulness and flow experience did show a small effect on learners' satisfaction. The effect size of perceived usefulness on continuance usage intention was also small. On the other hand, flow experience and learning self-efficacy in MOOCs had no effect on learners' satisfaction. Learners' satisfaction also did not possess any effect on continuance usage intention. The R^2 values were 0.517, 0.558, 0.680, 0.484 and 0.685, respectively. The explanatory variables could explain 51.7%, 55.8%, 68.0%, 48.4 and 68.5% of the variations in flow experience, perceived usefulness, learner's satisfaction, learning self-efficacy in MOOCs and continuance usage intention. In addition, all the Q^2 values were greater than zero, which indicated that exogenous constructs had predictive relevance for endogenous constructs.

Lastly, the effect and performance of expectation confirmation, flow experience, perceived usefulness, learners' satisfaction and learning self-efficacy in MOOCs that could impact continuance usage intention of MOOCs were verified via the importance-performance matrix analysis (IPMA). IPMA is an effective tool to identify those factors that are important but fail to perform well, and such factors deserve further managerial attention (Ringle and Sarstedt, 2016).

As indicated in Table 8 and Figure 4, the most important factor is expectation confirmation, while the least important factor is learners' satisfaction. On the other hand, the top performance was achieved in perceived usefulness (73.037), followed by learners'

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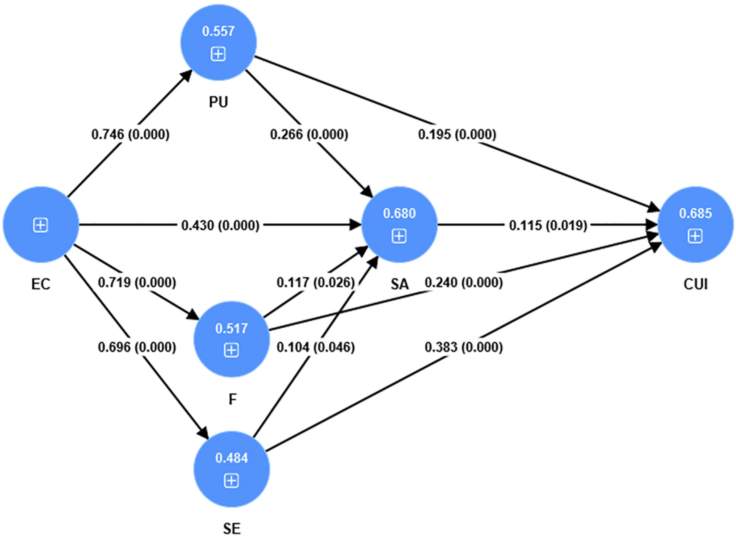

Source(s): Author's own work

Figure 3. Path coefficient and *p*-values of hypothesis testing

satisfaction (68.822), learning self-efficacy in MOOCs (68.072), expectation confirmation (67.733) and flow experience (65.979). The IPMA analysis reveals that although expectation confirmation is the core factor explaining continuance usage intention, it has not been performing satisfactorily. This result implies that MOOC providers fail to achieve expectation confirmation among the learners of MOOCs.

6. Discussion

The findings of this study indicate a noteworthy positive correlation between expectation confirmation and the satisfaction of learners. The result is consistent with the finding of Alraimi *et al.* (2015). When the learners' real performance surpasses their prior expectations on MOOC platforms, they feel satisfied with their MOOC learning. Expectation confirmation was found to have a positive relationship with perceived usefulness. The finding is consistent with that of Stone and Baker–Eveleth's study (2015). When learners' previous expectations are met, they will believe that the system is beneficial to use and will anticipate more from it in the future. Besides that, expectation confirmation is positively related to flow experience. The result is in line with that of Lu *et al.*'s (2019) study. When the learners' expectation

	Importance	Performance
Expectation confirmation	0.679	67.733
Learning self-efficacy in MOOCs	0.426	68.072
Flow experience	0.253	65.979
Perceived usefulness	0.233	73.037
Learners' satisfaction	0.119	68.822

Source(s): Created by author's

Table 8. Importance-performance matrix analysis

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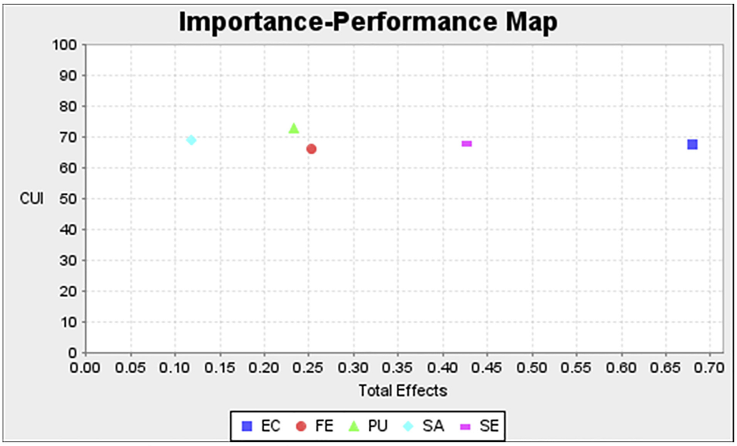

Figure 4. Importance-performance matrix analysis for continuance usage intention

Source(s): Author's own work

confirmation is achieved, they tend to have a greater eagerness and commitment to use MOOCs. As a result, a flow experience is created. Expectation confirmation also possesses a positive effect on self-efficacy. Gupta *et al.* (2020) support the finding that users' post-adoption self-efficacy, influenced by positive confirmation, is expected to play a crucial role in determining their increased confidence in exploring new features and functions of high-involvement technologies to achieve novel outcomes.

The study revealed a positive association between perceived usefulness and both learners' satisfaction and intention to continue usage. These results align with the findings of Gu *et al.* (2021). If the learners perceived that using MOOCs will improve their job or learning performance, this will increase their satisfaction and continuance usage intention. This study has also found that learners' satisfaction has a positive impact on continuance usage intention, which is also in consistent with that of Gu *et al.*'s (2021) study. This is because learners are more likely to continue using MOOCs when they feel satisfied with this type of learning platform. The present study found a positive relationship between flow experience and learners' satisfaction and continuous usage intention. These results are in line with that of Cheng's (2014) and Lu *et al.*'s (2019) studies, respectively. In sum, consumers' actions in virtual environments will engage flow experience with a variety of favorable effects such as satisfaction and behavioral intentions (Novak *et al.*, 2000).

Lastly, this study also indicates that learning self-efficacy in MOOCs has a positive impact on learners' satisfaction as well as continuance usage intention. The results are consistent with the findings of Heng *et al.* (2020). The idea is that when individuals feel that they can use technology to successfully achieve a task, they will engage in that work and put out all their efforts in it. By doing so, users who feel very satisfied after finishing the activity are more likely to have a stronger sense of self-efficacy (Agourram *et al.*, 2019). In addition, the ability of a person to utilize an electronic device to conduct and complete a proper task encourages repeated behaviors (Hsu and Chiu, 2004).

7. Conclusion

With the main aim of understanding and predicting learners' intention of engaging in MOOCs, this study was embarked to investigate the significant determinants of their continuance usage intention in this online platform. Based on the research results, the

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continuance usage intention has been found to be determined by perceived usefulness, learners' satisfaction, flow experience and learning self-efficacy. The identification of these influencing and motivating key factors will help providers to take the necessary measures to improve the completion rate of MOOCs, a unique online self-learning mode in the education industry of China. The present study also provides significant implications to other similar self-learning online platforms such as micro-credentials. The providers of these unique learning platforms will be able to strategize to attract and retain this unique learner market for sustainable performance.

7.1 Theoretical implications

This study has extended ECM by integrating two new constructs – flow and learning self-efficacy – in examining the factors affecting leaners' continuance usage intention toward MOOCs in China. In other words, ECM has been further extended with two other psychological theories, that is, flow theory and self-efficacy theory. By addressing this research gap in this empirical study, a new step has been put forward in terms of studying continuance usage intention in the context of MOOCs. Given the similarities between MOOCs and micro-credentials, these theoretical contributions can be extended to micro-credential or other similar learning platforms.

7.2 Practical implications

The findings of this research study are of great significance to the MOOC platform developers and course suppliers such as the higher learning institutions in China. The MOOC platform developers and course suppliers should try to enhance learners' expectation confirmation, especially at the beginning of the courses. The amount of confirmation or disconfirmation is determined by the gap between expectations (pre-usage) and perceived advantages (post-usage). Thus, MOOC platform developers and course suppliers should strategize to improve perceived benefits such as the academic achievement and learning efficiency to establish a positive expectation confirmation among the learners.

The MOOC providers should offer qualified or higher-quality courses which are the most fundamental yet important to the learners. These courses ensure the learners' long-term commitment and contentment. Furthermore, these courses should also tailor to various groups of learners with different levels and different needs. Different groups of learners will be able to find their own interesting and best-fitting courses to achieve their respective learning outcomes, academic goals and career development. Subsequently, MOOCs learners will respond with a higher perceived usefulness in MOOCs. They may become lifelong learners with a long-lasting passion for learning on MOOC platforms.

The MOOC platform providers may create opportunities for users to gain a greater flow experience by offering unique and creative course contents, services and functions. One will not be able to enjoy flow if he or she is being distracted without having the desire to continue learning (Nakamura *et al.*, 2009). To achieve this state, any unnecessary attention-grabbers in the platform must be removed. The MOOC developers should keep the webpage simple and neat without pop-up ads. They should also include teamwork functions like learners' online communities. These online communities allow learners to work as a team, seek help from one another and share their learning experience. Learners will experience flow when they find it much more enjoyable in team learning setting Walker's (2010) empirical study showed that learning in a social setting is much more enjoyable than solitary learning. The MOOC developers and course suppliers must think creatively to keep learners highly engaged in activities while learning MOOCs.

Besides upgrading the internet infrastructure, especially in the remote China countryside, MOOC providers may consider further improving on the MOOCs with user-friendly functions, interactivity, add on challenging course content, longer deadline for assignments

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and timely feedback mechanism from the lecturers. All these measures can be the contributing factors for motivating learners and improving their learning self-efficacy in MOOCs (O'Donovan *et al.*, 2013).

Due to the correlation between MOOCs and micro-credentials, the present investigation may benefit the practitioners of micro-credentials. The MOOC platforms would foster the development of micro-credentials. The findings of this study indirectly provide useful information to the micro-credential providers to enhance learners' continuance usage intention, which, in turn, would help improve the completion rate of such self-learning mode.

7.3 Research limitations and future research directions

The respondents selected in this study were college students in six universities of the first-tier cities of China, with just a very few of them being working adults. Future studies should expand the respondents to working adults, as this working group occupies 38% of the whole learner body, very close to the student group which constitutes 38.5% of the learning population (Zhiyan Consulting, 2016). Expanding the respondents of MOOCs to working adults also serves as a reference for the practitioners of micro-credentials to serve other learning needs of the working adults.

Previous studies have conceptualized flow experience into four dimensions: -focused attention, control, curiosity and temporal dissociation (Li and Browne, 2006). Similarly, self-efficacy has been conceptualized into three dimensions: -information searching, making queries and MOOC learning (Ghazali *et al.*, 2021). Future studies may investigate the separate effects of these dimensions in relation to continuance usage intention on e-learning platforms. This will provide a deeper understanding on how these two variables influence MOOCs learners' continuance usage intention.

Finally, this study only considers the internal factors (individual perceptual and experience factors) that affect the learners' continuous usage intention of MOOCs. Future studies may examine the impacts of external factors such as service quality (Ali *et al.*, 2022a, b) and authority's regulations in this online learning platform. Besides that, mediation analysis can be included in future study to provide more insights of the dynamic relationships of the variables.

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